Operations Research (M232)

Mock Midterm 2 — questions with complete solutions

100 points, two hours. Lectures 1 to 5. The paper first, then every part worked in full.

Problem 1 — Modelling and equational form (20 pts)

A workshop makes tables (x_1), chairs (x_2) and desks (x_3). A table sells for 90, a chair for 45, a desk for 70. One unit of wood costs: a table 10, a chair 4, a desk 7, and at most 260 units are available. Labour hours: 6, 3 and 5 per item, at most 180 hours. At most 20 chairs can be sold. A contract requires that tables plus desks total at least 10.

(a) Write the linear program that maximises revenue.

(b) Put the program in equational form $\max\{c^\top x : Ax = b,\ x \geq 0\}$, naming every slack and surplus variable.

(c) Consider the two bases $B_1 = \{s_1, s_2, s_3, s_4\}$ (all structural variables non-basic) and $B_2 = \{x_1, s_1, s_2, s_3\}$. Compute the corresponding basic solutions, say which ones are feasible, and give the objective value of each feasible one.

(d) How many candidate bases does the equational form have in total?

Problem 2 — Simplex run and optimality certificate (25 pts)

$$\max\ 5x_1 + 4x_2 \quad \text{s.t.} \quad 6x_1 + 4x_2 \leq 24, \qquad x_1 + 2x_2 \leq 6, \qquad x_1, x_2 \geq 0.$$

(a) Write the canonical form with slacks s_1, s_2 and the initial tableau (basis s_1, s_2), stating your reduced-cost convention.

(b) Run the simplex method to optimality. For each iteration give: the entering variable with its reduced cost, the ratio test, the leaving variable, and the full new tableau.

(c) Read the final tableau completely: optimal basis, optimal solution, optimal value, and the reduced costs of all four variables.

(d) Write the dual of this program, give its optimal solution, and verify complementary slackness in both directions.

Problem 3 — Duality and sign rules (22 pts)

Consider the program

$$\min\ 8x_1 + 6x_2 \quad \text{s.t.} \quad 2x_1 + x_2 \geq 3, \qquad x_1 + 2x_2 \leq 10, \qquad x_1 - x_2 = 2, \qquad x_1 \text{ free},\ x_2 \geq 0.$$

(a) Write its dual, justifying the sign of every dual variable and the direction of every dual constraint.

(b) The pair $x = (3, 1)$ and $y = (0, 0, 8)$ is made of a feasible primal point and a feasible dual point. Verify whether it satisfies complementary slackness, and conclude whether both can be optimal.

(c) Use weak duality to give a lower bound on the optimal primal value from the dual point of (b), and compare it with the primal objective value at x.

Problem 4 — Sensitivity analysis on the final tableau (23 pts)

You are given the optimal basis of Problem 2: $B = \{x_1, x_2\}$ with

$$B = \begin{bmatrix} 6 & 4 \\ 1 & 2 \end{bmatrix}, \qquad B^{-1} = \frac{1}{8}\begin{bmatrix} 2 & -4 \\ -1 & 6 \end{bmatrix}, \qquad x_B = \begin{pmatrix} 3 \\ 3/2 \end{pmatrix}, \qquad y^\top = (\tfrac{3}{4},\ \tfrac{1}{2}), \qquad z^* = 21.$$

(a) The right-hand side $b_1 = 24$ becomes $24 + \Delta$. Give $x_B(\Delta)$, the range of Δ for which the basis stays feasible, and the optimal value on that range. What is the shadow price interpretation of the coefficient of Δ?

(b) The profit $c_1 = 5$ of the basic variable x_1 becomes $5 + \theta$. Give the range of θ for which the basis stays optimal.

(c) A new activity is proposed: it consumes 3 units of the first resource and 1 of the second, and earns 4. Does it enter the basis? Justify with a reduced-cost computation.

(d) True or false, justified in one sentence: at this optimal basis, if some basic variable had value 0, the vertex would be degenerate and a pivot could leave the objective unchanged.

Problem 5 — True or false (10 pts, justify each)

1. Every linear program in standard form that has a feasible solution also has a basic feasible solution.
2. If the primal minimisation program is unbounded below, then the dual is infeasible.
3. A degenerate basic feasible solution can be represented by several different bases.
4. For a maximisation program in canonical form, an optimal basis has all reduced costs $c_j - z_j \leq 0$.
5. The dual simplex method starts from a basis that is primal feasible and dual infeasible.

Solutions

Problem 1

Solution – a

$$\begin{aligned} \max \quad & 90x_1 + 45x_2 + 70x_3 \\ \text{s.t.} \quad & 10x_1 + 4x_2 + 7x_3 \le 260 \quad \text{(wood)} \\ & 6x_1 + 3x_2 + 5x_3 \le 180 \quad \text{(labour)} \\ & x_2 \le 20 \quad \text{(chair demand)} \\ & x_1 + x_3 \ge 10 \quad \text{(contract)} \\ & x_1, x_2, x_3 \ge 0. \end{aligned}$$

Solution – b

Slacks $s_1, s_2, s_3 \ge 0$ for the three $\le$ rows, surplus $s_4 \ge 0$ for the $\ge$ row:

$$\begin{array}{rcrcrcrcrcl} 10x_1 & + & 4x_2 & + & 7x_3 & + & s_1 & & & = & 260 \\ 6x_1 & + & 3x_2 & + & 5x_3 & & + & s_2 & & = & 180 \\ & & x_2 & & & + & & s_3 & & = & 20 \\ x_1 & & & + & x_3 & & & & - & s_4 & = & 10 \end{array} \qquad x, s \ge 0, \quad \max\ 90x_1 + 45x_2 + 70x_3.$$

Solution – c

$B_1 = \{s_1, s_2, s_3, s_4\}$: set $x_1 = x_2 = x_3 = 0$, then $s_4 = -10 < 0$ from the fourth row. The basic solution is $(0, 0, 0\,;\, 260, 180, 20, -10)$: **infeasible** (the surplus cannot absorb the contract).
$B_2 = \{x_1, s_1, s_2, s_3\}$: set $x_2 = x_3 = s_4 = 0$, then $x_1 = 10$, and $s_1 = 260 - 10 \cdot 10 = 160$, $s_2 = 180 - 6 \cdot 10 = 120$, $s_3 = 20$. All non-negative, so the solution $(10, 0, 0\,;\, 160, 120, 20, 0)$ is **feasible**, with objective $90 \cdot 10 = 900$.

Solution – d

Seven variables $(x_1, x_2, x_3, s_1, s_2, s_3, s_4)$, four equations: $\binom{7}{4} = 35$ candidate bases (some infeasible, some possibly singular, but 35 is the count of subsets).

Problem 2

Solution – a

Canonical form: max $5x_1 + 4x_2$ with $6x_1 + 4x_2 + s_1 = 24$, $x_1 + 2x_2 + s_2 = 6$, all variables ≥ 0. Convention: the objective row displays the reduced costs $c_j - z_j$; a *positive* entry improves a maximisation.

	x_1	x_2	s_1	s_2	
s_1	6	4	1	0	24
s_2	1	2	0	1	6
z	5	4	0	0	0

Solution – b

Iteration 1. Entering x_1 (reduced cost 5, the largest). Ratios: $24/6 = 4$ and $6/1 = 6$; minimum

4, so s_1 leaves. Pivot on the 6 (row of s_1):

	x_1	x_2	s_1	s_2	
x_1	1	$\frac{2}{3}$	$\frac{1}{6}$	0	4
s_2	0	$\frac{4}{3}$	$-\frac{1}{6}$	1	2
z	0	$\frac{2}{3}$	$-\frac{5}{6}$	0	20

Iteration 2. Entering x_2 (reduced cost $\frac{2}{3}$). Ratios: $4/\frac{2}{3} = 6$ and $2/\frac{4}{3} = \frac{3}{2}$; minimum $\frac{3}{2}$, so s_2 leaves. Pivot on $\frac{4}{3}$:

	x_1	x_2	s_1	s_2	
x_1	1	0	$\frac{1}{4}$	$-\frac{1}{2}$	3
x_2	0	1	$-\frac{1}{8}$	$\frac{3}{4}$	$\frac{3}{2}$
z	0	0	$-\frac{3}{4}$	$-\frac{1}{2}$	21

All reduced costs are ≤ 0: stop.

Solution – c

Optimal basis $B = \{x_1, x_2\}$, optimal solution $x_1 = 3$, $x_2 = \frac{3}{2}$, $s_1 = s_2 = 0$, optimal value $z^* = 21$. Reduced costs: x_1 and x_2 basic (0), s_1: $-\frac{3}{4}$, s_2: $-\frac{1}{2}$. No positive reduced cost certifies optimality.

Solution – d

Dual: min $24y_1 + 6y_2$ s.t. $6y_1 + y_2 \geq 5$, $4y_1 + 2y_2 \geq 4$, $y \geq 0$. From the final tableau, $y^\top = c_B^\top B^{-1} = (5,4)\frac{1}{8}\begin{bmatrix} 2 & -4 \\ -1 & 6 \end{bmatrix} = (\frac{3}{4}, \frac{1}{2})$, with objective $24 \cdot \frac{3}{4} + 6 \cdot \frac{1}{2} = 18 + 3 = 21 = z^*$ (strong duality).
Complementary slackness, primal side: $y_1 = \frac{3}{4} > 0 \Rightarrow 6x_1 + 4x_2 = 24$ (tight: $18 + 6 = 24$); $y_2 = \frac{1}{2} > 0 \Rightarrow x_1 + 2x_2 = 6$ (tight: $3 + 3 = 6$). Dual side: $x_1 = 3 > 0 \Rightarrow 6y_1 + y_2 = 5$ (tight: $\frac{18}{4} + \frac{1}{2} = 5$); $x_2 = \frac{3}{2} > 0 \Rightarrow 4y_1 + 2y_2 = 4$ (tight: $3 + 1 = 4$). All four conditions hold: the pair is optimal.

Problem 3

Solution – a

Rules used: a $\geq$ row of a *minimisation* primal gives $y_i \geq 0$; a $\leq$ row gives $y_i \leq 0$; an equality gives y_i free. A free primal variable gives a dual *equality*; a non-negative primal variable gives a dual $\leq$ constraint (against the minimisation cost vector).

max $3y_1 + 10y_2 + 2y_3$ s.t. $2y_1 + y_2 + y_3 = 8$, $y_1 + 2y_2 - y_3 \leq 6$, $y_1 \geq 0$, $y_2 \leq 0$, y_3 free.

Solution – b

Feasibility: $x = (3,1)$: $2 \cdot 3 + 1 = 7 \geq 3$, $3 + 2 = 5 \leq 10$, $3 - 1 = 2$. $y = (0,0,8)$: $2 \cdot 0 + 0 + 8 = 8$, $0 - 8 = -8 \leq 6$, signs .
Complementary slackness: rows 1 and 2 are slack at x (gaps $7 - 3 = 4$ and $10 - 5 = 5$), so optimality would force $y_1 = y_2 = 0$ (true here); $x_2 = 1 > 0$ would force the second dual constraint *tight*: $y_1 + 2y_2 - y_3 = 6$, but $-8 \neq 6$. **CS fails**, so the pair cannot be simultaneously optimal. Indeed $c^\top x = 30$ and $b^\top y = 16$ differ.

Solution – c

Weak duality: for every feasible primal x and dual y, $b^\top y \le c^\top x$. The dual point gives $16 \le p^*$ (a lower bound on the optimal primal value). The primal point gives $p^* \le 30$. The gap $16 \le p^* \le 30$ shows both endpoints are merely bounds; since CS failed in (b), neither bound is tight.

Problem 4

Solution – a

$x_B(\Delta) = B^{-1}(b + \Delta e_1) = B^{-1}b + \Delta B^{-1}e_1 = \binom{3}{3/2} + \Delta\binom{1/4}{-1/8} = \binom{3+\Delta/4}{3/2-\Delta/8}$. Feasibility of the basis: $3 + \Delta/4 \ge 0 \Rightarrow \Delta \ge -12$ and $3/2 - \Delta/8 \ge 0 \Rightarrow \Delta \le 12$, so $\Delta \in [-12, 12]$. On that range $z(\Delta) = 5(3 + \frac{\Delta}{4}) + 4(\frac{3}{2} - \frac{\Delta}{8}) = 21 + \frac{3}{4}\Delta$. The coefficient $\frac{3}{4} = y_1$ is the shadow price of the first resource: each extra unit of b_1 raises the revenue by 0.75, valid while Δ stays in the range; beyond it the basis changes and the price no longer holds.

Solution – b

$c_B = (5+\theta, 4)$, so $y(\theta)^\top = c_B^\top B^{-1} = \frac{1}{8}(2(5+\theta) - 4,\ -(5+\theta) + 24) = \frac{1}{8}(6 + 2\theta,\ 19 - \theta - 15)$, i.e. $y_1 = \frac{6+2\theta}{8}$, $y_2 = \frac{4-4\theta}{8}$. Optimality keeps $y \ge 0$: $6 + 2\theta \ge 0 \Rightarrow \theta \ge -3$ and $4 - 4\theta \ge 0 \Rightarrow \theta \le 1$. So $\theta \in [-3, 1]$, i.e. $c_1 \in [2, 6]$: below 2 or above 6 the reduced cost of a slack turns positive and the basis must change.

Solution – c

Reduced cost of the new activity with column $a = \binom{3}{1}$ and profit 4:

$$4 - y^\top a = 4 - (\tfrac{3}{4} \cdot 3 + \tfrac{1}{2} \cdot 1) = 4 - \tfrac{11}{4} = \tfrac{5}{4} = 1.25 > 0.$$

A positive reduced cost at a maximisation optimum means the activity is **attractive**: it enters the basis and the current solution is no longer optimal. (Equivalently: it earns 4 but its opportunity cost through the two resources is only 2.75.)

Solution – d

True. A basic variable at value 0 is the definition of a degenerate vertex; a pivot on a row whose right-hand side is 0 does not move the point, so the objective stays unchanged even though the basis changes (this is also why cycling rules exist).

Problem 5

Solution – 1–5

1. True — fundamental theorem of LP: a non-empty feasible polyhedron of a standard-form program has at least one basic feasible solution (an extreme point).
2. True — if the dual had a feasible point, weak duality would bound the primal from below, contradicting unboundedness; so the dual is infeasible.
3. True — degeneracy means more than n hyperplanes through the point, so several bases represent the same vertex (as in the 35 candidates of Problem 1, several can collapse to one point).
4. True — for a max problem, $c_j - z_j \le 0$ for all j is exactly the optimality certificate; a positive reduced cost offers an improving direction.
5. False — it is the reverse: the dual simplex starts from a basis that is *dual* feasible (reduced costs fine) but *primal* infeasible (negative right-hand side), and repairs feasibility while keeping

optimality of the reduced costs.